

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Engineered **Spring Boot microservices** orchestrating **LLM-powered agents** using **OpenAI specs** and **Claude model integrations**, implementing **tool execution layers**, **function calling**, and **retry mechanisms** with guardrails for robust agentic workflows; achieved **99.2% agent execution reliability** in production environments.
- Implemented **MCP (Model Context Protocol) clients and servers** with structured **tool schema definitions** and **function orchestration** for developer-focused automations; integrated **GitHub APIs**, **GitHub Actions webhooks**, and **Cline workflows** to automate PR review and code generation pipelines, reducing manual review overhead by **40%**.
- Built **REST APIs and event-driven pipelines** for **agent request orchestration**, **context assembly**, **RAG (Retrieval-Augmented Generation)** with embeddings, and **result streaming**; implemented **prompt injection defenses**, **input validation**, and **secret management** guardrails to ensure compliance with AI security standards.
- Developed **observability infrastructure** with structured **logs**, **metrics**, **distributed traces**, and **run metadata dashboards** using **OpenTelemetry** and **Datadog**; tracked agent performance, latency, cost-per-request, and reliability SLAs across **20+ concurrent agentic services**.
- Designed and executed **comprehensive test coverage** including **unit tests**, **integration tests**, and **end-to-end agent scenario tests**; automated **CI/CD pipelines** with **Jenkins** and **GitHub Actions**, delivering zero-downtime deployments and maintaining **95%+ code quality standards** across production microservices.
- Collaborated across **front-end and platform teams** to define **service contracts**, **API specifications**, and **agentic workflow requirements**; led **design reviews** and **code reviews** for **LLM integration patterns**, **configuration management**, and **access control policies** aligned with least-privilege security architecture.

SKILLS

Backend & Microservices – Java, Spring Boot, REST APIs, Spring Cloud, microservices architecture, event-driven design.

LLM & AI Agent Engineering – OpenAI API, Claude (Anthropic SDK), LLM orchestration, agentic workflows, tool/function calling, MCP (Model Context Protocol), MCP clients and servers, prompt engineering, RAG, embeddings, vector stores.

Developer Tools & Integration – GitHub APIs, GitHub Actions, Cline, webhook orchestration, CI/CD automation, API integrations.

Security & Compliance – prompt injection defenses, input validation, secret management, least-privilege access control, AI service security, regulatory compliance.

Observability & Performance – OpenTelemetry, Datadog, structured logging, distributed tracing, metrics dashboards, performance tuning, SLA management.

Data & APIs – GraphQL, message/event systems (Kafka, SQS, AWS Lambda), data modeling, PostgreSQL, MySQL, Oracle, AWS (EC2, S3, Lambda, CloudWatch).

Testing & DevOps – unit testing, integration testing, JUnit, Mockito, Jenkins, Docker, Git, cloud deployment patterns, zero-downtime deployments.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Ireland Jobs Tracker — AI-Powered Resume Orchestration Engine (Personal Project)

- Built production-grade **LLM orchestration service** integrating **Claude API (Anthropic SDK)** with **structured JSON prompt engineering** and **constraint guardrails**; engineered end-to-end **agentic workflow automation** pipeline: job description ingestion → **Claude-powered content generation** → PDF synthesis → deployment.
- Implemented **tool execution framework** with **function calling** for dynamic JD analysis, resume tailoring logic, and structured output validation; optimized **token usage** and **cost-per-request** through intelligent caching and **Claude Haiku model routing**.
- Orchestrated **event-driven data pipelines** using **Apify scrapers** for job ingestion, **GitHub Actions CI** for workflow automation, and **Vercel deployment**; demonstrated mastery of **API integration**, **CI/CD automation**, and **cloud deployment patterns** directly applicable to eBay's agentic services.

SWIFT Payments Agent & Operations Dashboard (HCL Technologies)

- Engineered **Spring Boot REST APIs** for real-time **SWIFT message orchestration**, **function calling workflows**, and **tool execution** logic supporting **MT/MX message formats** and **ISO 20022 message flows**; implemented **guardrails** for payment validation and regulatory compliance (GDPR, PSD2, SEPA standards).
- Built **observability dashboards** for agent-driven payment operations using **Splunk**, **CloudWatch**, and **Dynatrace**; tracked **agent execution metrics**, **function call latency**, **error rates**, and **SLA compliance** across **10K+ daily transactions** with **99.5% reliability**.
- Implemented **event-driven pipeline** for cross-system payment orchestration via **REST APIs** and **message queues**; collaborated on **design reviews** and **service contracts** with banking operations and compliance teams, ensuring **audit trail integrity** and **least-privilege access controls**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)